Technical strategy

TMP Revamp & Demand Planning

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TMP Reimagination

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TMP Reimagination

Business Executive Summary

WHY: The Challenge

CURRENT CONSTRAINTS

Current systems inhibit agility and cannot support the required scale for future growth.

STRATEGIC GAP

Lack of democratized Talent Intelligence limits real-time adaptability to business needs.

WHAT: The Solution

Next-Gen Talent Marketplace

A mobile-enabled, AI-driven platform replacing legacy architecture with a democratized intelligence hub.

Key Personas Enabled

- Associate
- Hiring Manager
- TSC

Tech Core

Hyper-infusion of AI & Analytics for intelligent decision-making.

OUTCOME: Value

- **Efficiency**
  Automation @ Scale to unlock productivity and reduce manual effort.
- **Agility**
  Configurable architecture architected to support growth.
- **Experience**
  Frictionless, intuitive journeys, agentic for all key personas.
- **Compliance & Insight**
  Improved predictability and governance via data analytics.

EA ASK: Approval

We request Architecture approval for the proposed:

- Technical Architecture
- Technology Stack

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TMP Reimagination

Technical Executive Summary

MVP Goal & Approach

- **Reuse & Modernize:** Leverage current business logic services wherever needed while enhancing UX with modern look along with agentification.
- **Targeted Enhancement**: Selectively uplift services; keep stable core AS-IS services untouched.
- **Accelerate Value**: Introduce agents for rapid automation wins without full rewrite.
- **Architecture Standards**: Additional features will be built using the same tech stack by adhering to and uplifting architecture standards on top of current implementation

Service based architecture -> Microservices architectural style.

FOCUSED ON 3 KEY PERSONAS

- Associate
- TSC
- Hiring Mgr

Platform Coexistence

Tech Strategy

Reuse: Resuse existing WCF Services , ,Net cire services expose via API Gateway

Enhance: Enhance by wrapping logic via new .NET Core API

New: Build new Micro services with **.NET core** by following MS architectural style and **Python** AI Agent Layer integration and & **React** for UI.

CORE (AS-IS)

.NET | Azure | SQL | SSIS

MODERN (NEW)

Python | Autogen | React | GCP | Micro services architecture | Postgres SQL

- Azure + GCP Coexistence
- Service Based - Domain Micro Services
  - Shared DB
  - Domain Microservices
- End Goal: Postgres
- End Goal: Full GCP Migration

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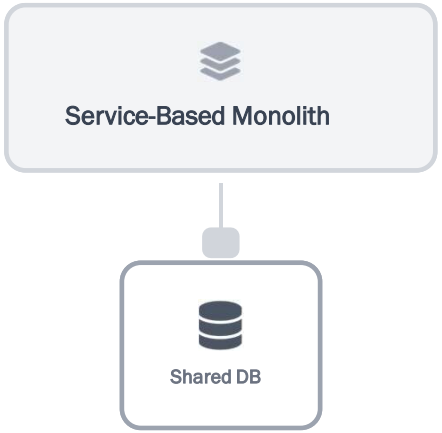
Architecture Transformation

Structural Evolution: Monolith to Microservices

A Current State

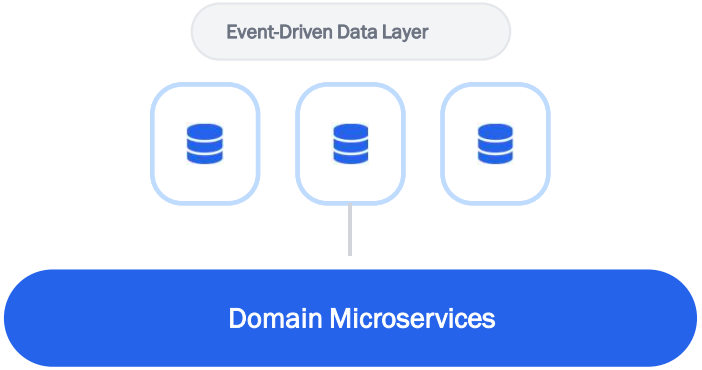
CURRENT



Single logical executable with shared data persistence, creating tight coupling and deployment dependencies.

B Final Target State

TARGET



Eventual move to domain databases + event-driven ETL for decoupled, scalable architecture.

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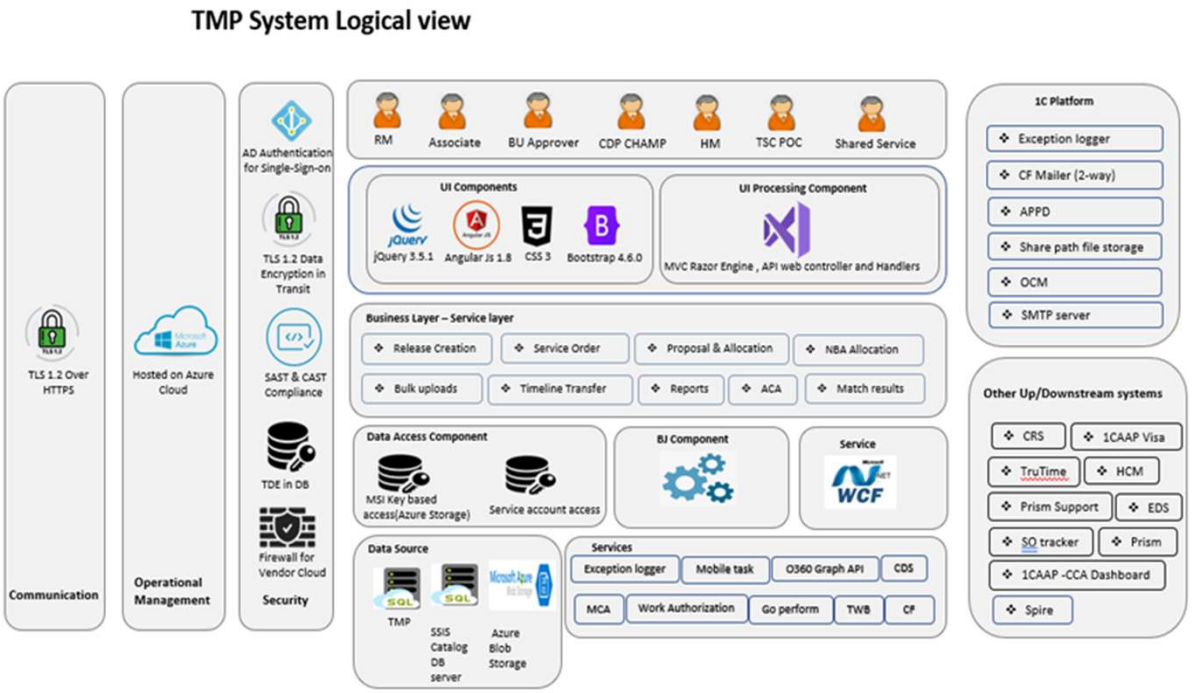
AS-IS Architecture Overview



UI & Authentication

Access: AD Authentication (Single-Sign-On)

Processing: MVC Razor Engine, API Web Controllers

Frameworks (current): jQuery 3.5.1 Angular 2.1.8 Bootstrap 4.0

Business Logic & Services

Core Modules: Release Creation, Proposal & Allocation, NBA Allocation, BQI Uploads, Reports, ICA.

Service Layer: Monolithic WCF Services, Timeline Transfers.

Up/Downstream: 15+ Integrations

Data & Security

Storage: Azure Blob Storage, SSS Catalog DB, TMP Data Source.

Security: TLS 1.2 over HTTPS, TDE in DB, Firewall for Vendor Cloud, Key-based access.

Ops: Exception Logger, CF Mailer, SMTP Server.

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AS-IS Challenges & Pain Points

Current Tech Stack

Older Tech: Reliance on older jQuery 3.5.1 and Angular 2.1.8 limits UI capabilities.

Uneven UX: Inconsistent user experience across modules due to mixed legacy technologies.

Limited Automation

Monolithic (Service Based) Structure

Tight Coupling: Intertwined business logic makes isolated updates risky. Following Service based architecture.

Slow Cycles: Full regression needed for minor changes, slowing release cadence.

Scalability: Cannot scale individual services

Complex Integrations

Brittle Mesh: Point-to-point WCF services create a fragile dependency chain.

Dependency : 15+ external systems (HCM, EDS, Prism) with tight coupling.

Failures: High failure rate during upstream/downstream updates.

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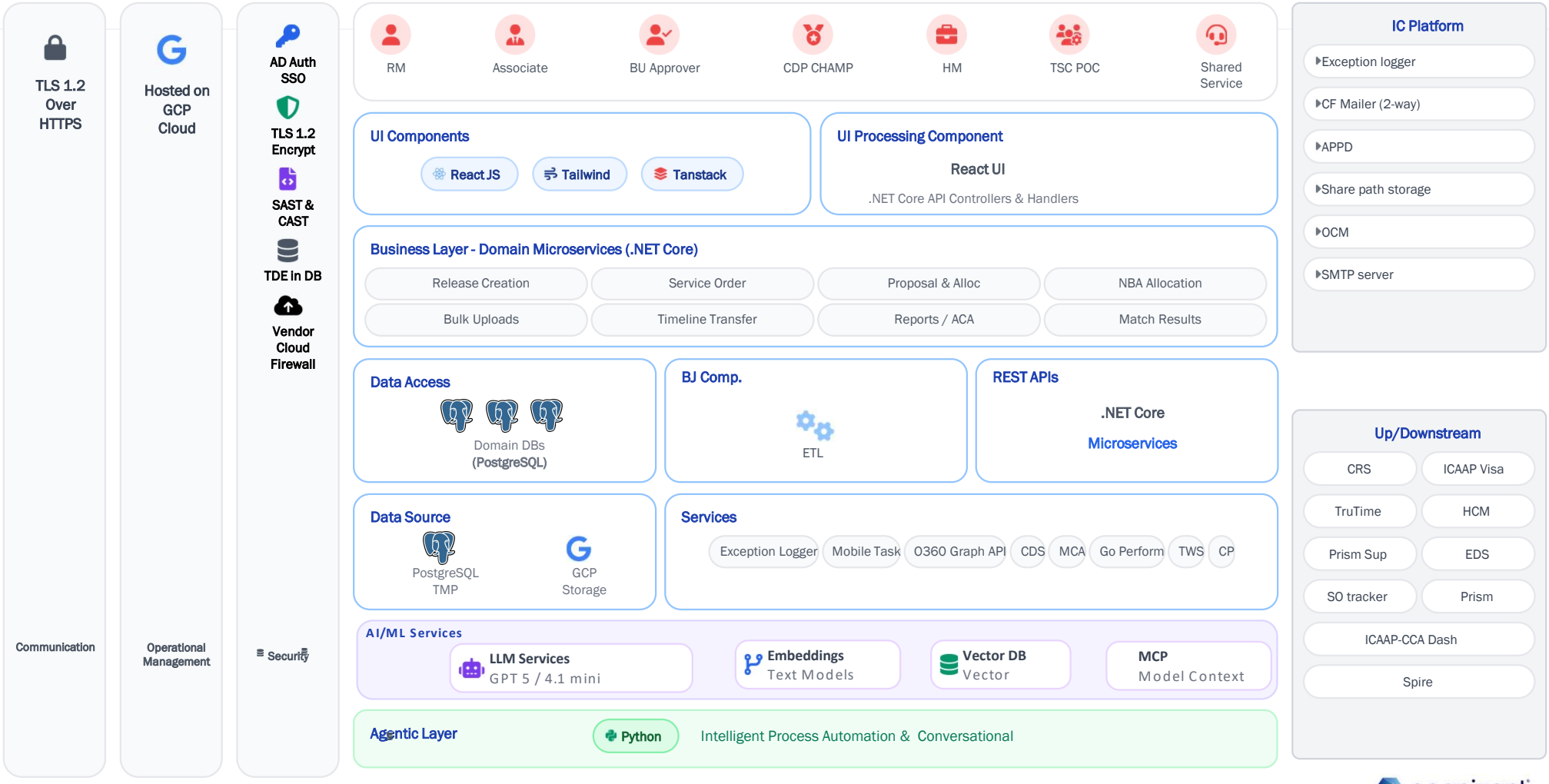
TO-BE Architecture Overview Logical View



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Architectural Strategy & Decisions

Strategic Roadmap & Infrastructure Planning

Architecture Overview

Modernization Vision & Key Drivers

The proposed solution represents a fundamental shift from a legacy monolithic structure to a Cloud-Native, Domain-Driven Microservices Architecture.

SCALABILITY

Independent scaling of critical talent supply chain functions.

DEMOCRATIZATION

Data access via a standardized API layer across the enterprise.

INTELLIGENCE

Agentic AI layer for intelligent process automation.

Hybrid Cloud Transition Strategy

Transition Phase & MVP Deployment

- Azure + GCP Coexistence

  Both cloud platforms will run in parallel during Transition & MVP phases to ensure business continuity.

- End Goal: Full GCP Migration

  Strategic objective to eventually migrate all workloads to Google Cloud Platform as the primary target environment.

PENDING DECISION: DATA LAYER

Placement: Azure SQL vs. PostgreSQL on GCP

Under active discussion due to complexity of SSIS packages, high data volumes, table dependencies, and migration effort required.

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Technology Migration Path (MVP)

AS-IS State Current PLATFORM		TO-BE State
Current UI Angular 2.x / jQuery	COEXIST / REWRITE	**Modern Experience** React + Conversational UI
Core Business Logic WCF Services & .Net Services (Reused) Service based architecture	REUSE & ENHANCE	**Enhanced Services** ASP.NET Core API Wrappers **New Services** ASP .NET core Micro Services Architecture
No Agent Capability Manual Workflows	NEW PYTHON LAYER	Python **Agent Layer** LangChain / AutoGen Reuse Existing agents/workflows
SQL Server Centralized Data	RETAIN DATA	**SQL + Vector DB** Data Coexistence During Transition Stage → **Postgres SQL + Vector DB** Data Coexistence
Azure Infrastructure Existing Landing Zone	SHARED INFRA	**Azure Infrastructure** During Transition Stage → **GCP Infrastructure**

● Reuse / Maintain ● Enhance / Wrap ● New (Python) ● Modernize

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Resources provision and configuration for Development - TMP

Stack	Configuration	Cloud Provider/Framework
LLM	Gemini 3 Pro, Gemini 2.5 Flash	GCP
Embedding Model	All-mpnet-base-v2	Hugging Face
Google Cloud Storage	200 GB initial capacity; Hot tier; Hierarchical namespace	GCP
Vector DB	Deploy Qdrant on Google Kubernetes Engine Cluster using a docker image	GCP
Postgres DB	Google Cloud's managed service for PostgreSQL Flexible Server, 4 vCores, 128-256 GB storage (auto-grow)	GCP
During Transition Stage SQL DB	Azure SQL Database	Azure
During Transition Stage Azure DevOps (ADO)	Repos + Pipelines + Environments; Service connection to Azure	Azure
GCP Deployment Related	Repos + Pipelines + Environments; Service connection to GCP	GCP

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Demand Planning

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Demand Planning Program

Forecasting Transformation Strategy

PRIMARY OBJECTIVE

Transform demand forecasting into automated, accurate processes to drive revenue growth, cost efficiency, & faster talent fulfillment.

Program Objectives

Hire to Plan
Shift from ad-hoc account-based hiring to planned, data-backed hiring decisions.

Granular Forecasting
Forecast and act at leaf-skill level for greater accuracy; roll up to hierarchical level & aggregate versions.

Aligned Framework
Establish a single, aligned demand framework with world class ML Models for integrating account, Role, Location, skill, and time dimensions (60-180 days).

Business Impact

Revenue & Cost
Revenue uplift and cost savings through better staffing alignment and win realization.

Confidence Ratio
Increase % of demand backed by confirmed projects.

Fulfillment Speed
Faster response times for fulfillment (bench, rotations, sub-con, visa readiness).

Forecast Accuracy
Reduce deviation between forecasted and actual demand at leaf level skills & roles.

Key Users

Service Line Teams
COO, Practice Heads

Market Teams
COO, SBU Heads

Back End Team

Platform Mgmt
MLOps, Data Eng.

Central Forecast
Admin, Planning

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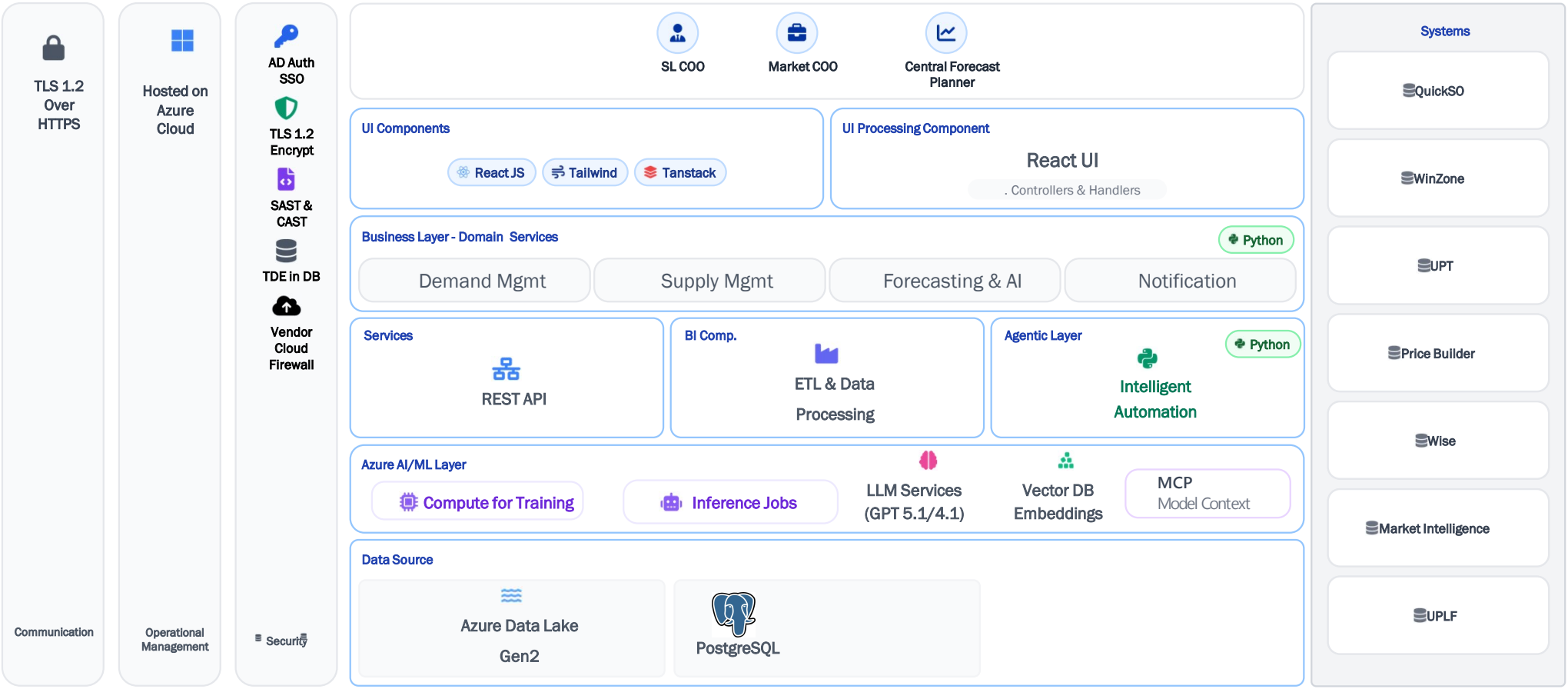
Demand Planning Technical Architecture

Logical View



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Architectural Overview

Architecture Overview

Azure-Native • Domain-Driven • Intelligent

The reimagined platform leverages an Azure-native, domain-driven design powered by Python Services. This architecture shifts from legacy monoliths to agile, independent services, hyper-infused with AI/ML capabilities through a dedicated Agentic Layer and modern REST API modernization.

Key Technology Highlights

Core Stack Components & Rationale

Python Domain Services

Leveraging Python's rich ecosystem for backend logic, enabling seamless AI/ML integration and rapid development cycles.

REST API Modernization

Standardized, stateless communication layer ensuring interoperability and scalable service consumption.

Azure ML & Agentic Layer

Dedicated compute for training and inference jobs, enabling intelligent process automation and orchestration.

Azure Data Lake Gen2 & ETL

Scalable storage for massive datasets with modernized ETL pipelines for real-time data processing.

GenAI Integration (LLM)

Integration of GPT 5/4.1 models and text embedding models for advanced language tasks and cognitive capabilities.

Vector DB

Specialized storage for vector embeddings to support semantic search and RAG architectures.

Strategic Benefits

Value Realization

Azure Scalability

Elastic infrastructure adapting to demand spikes.

Python Ecosystem

Access to vast libraries for data science & backend.

AI-Native

Architecture built primarily for AI/ML workloads.

Integration Ecosystem

Connected Data Systems

QuickSO | WinZone | UPT | Price Builder | Wise

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Technology Stack

Tech Stack

Modern Experience
React + Conversational UI

New Services
Python

Agent Layer Python
LangChain / AutoGen

Postgres SQL + Vector DB
Data

Azure Infrastructure

Stack	Configuration	Access details
LLM (Azure OpenAI)	GPT Models – 5.1 mini and 4.1 mini. Input Tokens : 15 Million, Output Tokens: 5 Million (for now).	Access required from both Workstation and VM
Embedding Model	text-embedding-small-3	Access required from both Workstation and VM
Data Lake Gen2	200 GB initial capacity; Hot tier; Hierarchical namespace	Access required from both Workstation and VM
Vector DB	Expecting Qdrant Managed Service (Currently we got pgvector)	Access required from both Workstation and VM
Postgres	Azure Database for PostgreSQL Flexible Server, 4 vCores, 128-256 GB storage (auto-grow)	Access required from both Workstation and VM
Azure DevOps (ADO)	Repos + Pipelines + Environments; Service connection to Azure	
MSI keys	System-assigned on AML compute and endpoints	
Container Registry	Azure Container	

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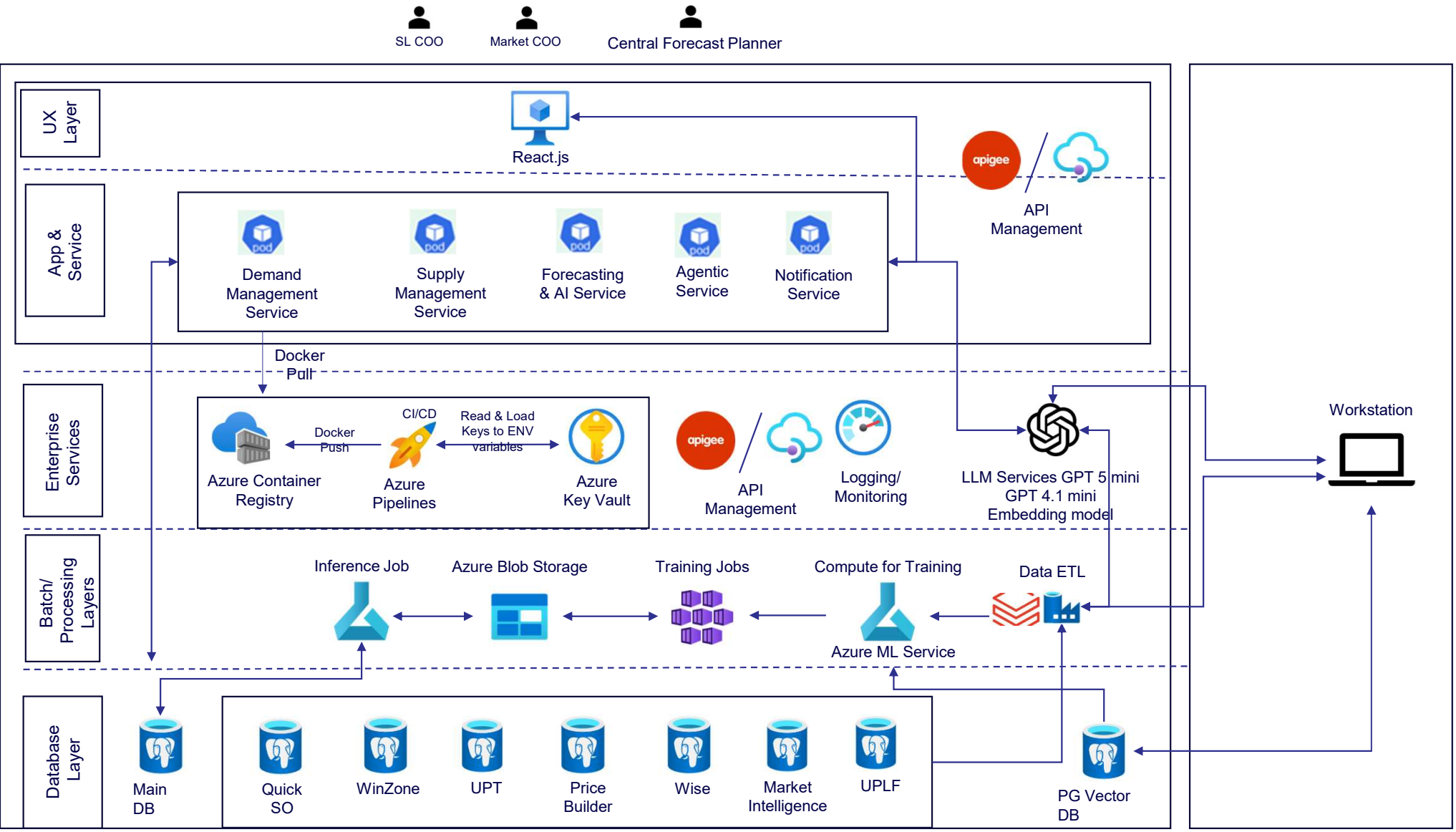
SL COO
Market COO
Central Forecast Planner
UX Layer
React.js
API Management
App & Service
Demand Management Service
Supply Management Service
Forecasting & AI Service
Agentic Service
Notification Service
Docker Pull
Enterprise Services
Azure Container Registry
Docker Push
CI/CD
Azure Pipelines
Read & Load Keys to ENV variables
Azure Key Vault
API Management
Logging/ Monitoring
LLM Services GPT 5 mini
GPT 4.1 mini
Embedding model
Workstation
Batch/ Processing Layers
Inference Job
Azure Blob Storage
Training Jobs
Compute for Training
Data ETL
Azure ML Service
Database Layer
Main DB
Quick SO
WinZone
UPT
Price Builder
Wise
Market Intelligence
UPLF
PG Vector DB

Why OpenAI GPT-5.1 Instead of Opensource ?

Native Support for Agentic Actions

Built-in function calling capabilities for tools integration and seamless agent workflows

JSON Mode & Structured Output

Guaranteed parseable responses for reliable system integrations and API-driven workflows

Advanced Reasoning & Accuracy

Superior performance on complex tasks with lower hallucination rates than open alternatives

Enterprise-Grade Reliability

Production-ready infrastructure with guaranteed SLAs, compliance, and security controls

Cost vs. Effort Trade-off

Lower total cost of ownership compared to building, tuning, and maintaining open source LLMs

Easy Ecosystem Integration

Seamless connectivity with Azure AI services, security, and governance frameworks

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Why Qdrant

Feature	Qdrant	Pgvector
Purpose-built vector DB	Native vector database optimized for similarity search	Postgres extension for storing vectors
Indexing options	HNSW index with quantization, optimized for high-dimensional search	Basic flat index, limited ANN support
Hybrid search	Combines vector + keyword filtering seamlessly	Requires manual SQL joins
Payload filtering	Rich metadata filtering (e.g., tags, categories)	Limited filtering
Distributed clustering	Built-in horizontal scaling and replication	Scaling depends on Postgres setup, not native
Performance	Optimized for low-latency vector queries	Slower at scale, especially with millions of vectors
Developer experience	REST/gRPC APIs, SDKs, easy integration with AI pipelines	SQL-only, requires schema design and tuning
Operational tooling	Monitoring, observability, production-ready deployment	Relies on Postgres ecosystem, not vector-specific
RAG/AI workflows	Native support for retrieval-augmented generation	Needs custom orchestration
Community & ecosystem	Active vector DB community, AI-first integrations	General Postgres community, not vector-focused

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Thank you